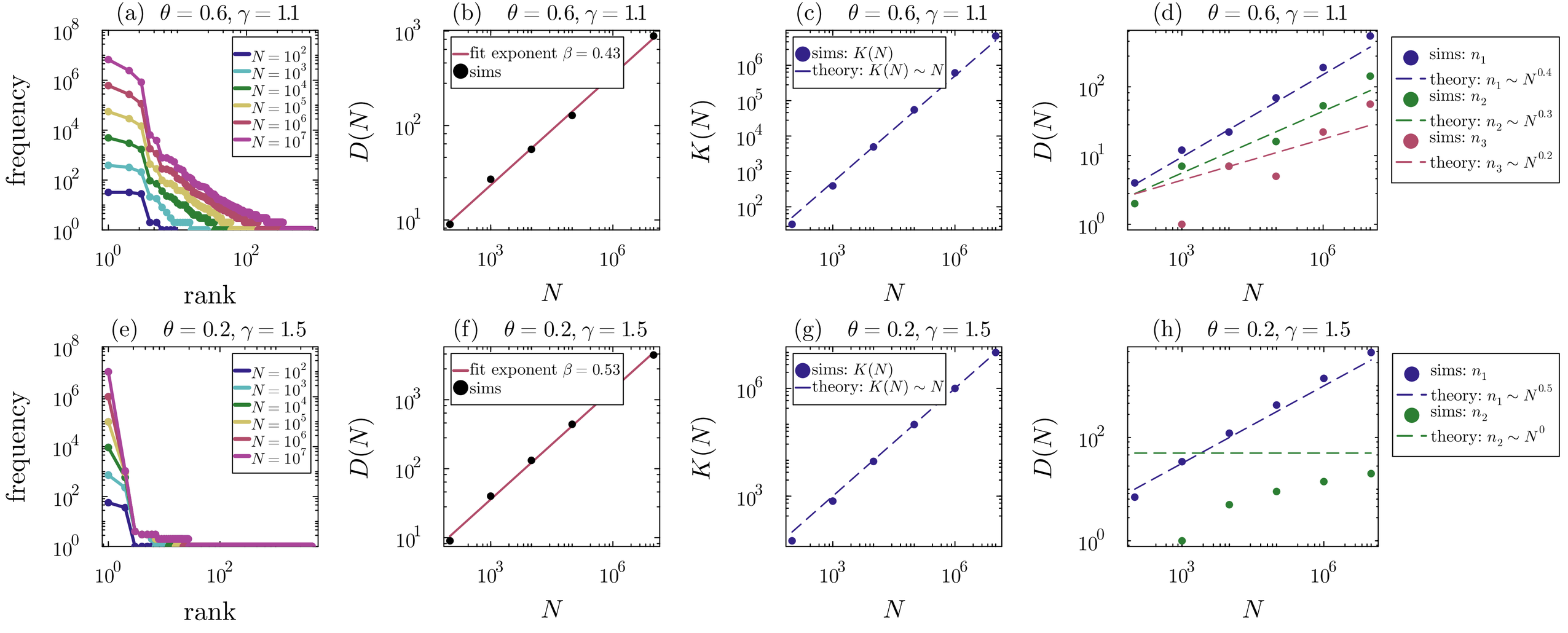

(a) $\theta = 0.6, \gamma = 1.1$
frequency
rank
$N = 10^2$
$N = 10^3$
$N = 10^4$
$N = 10^5$
$N = 10^6$
$N = 10^7$
(b) $\theta = 0.6, \gamma = 1.1$
$D(N)$
N
fit exponent $\beta = 0.43$
sims
(c) $\theta = 0.6, \gamma = 1.1$
$K(N)$
N
sims: $K(N)$
theory: $K(N) \sim N$
(d) $\theta = 0.6, \gamma = 1.1$
$D(N)$
N
sims: n_1
theory: $n_1 \sim N^{0.4}$
sims: n_2
theory: $n_2 \sim N^{0.3}$
sims: n_3
theory: $n_3 \sim N^{0.2}$
(e) $\theta = 0.2, \gamma = 1.5$
frequency
rank
$N = 10^2$
$N = 10^3$
$N = 10^4$
$N = 10^5$
$N = 10^6$
$N = 10^7$
(f) $\theta = 0.2, \gamma = 1.5$
$D(N)$
N
fit exponent $\beta = 0.53$
sims
(g) $\theta = 0.2, \gamma = 1.5$
$K(N)$
N
sims: $K(N)$
theory: $K(N) \sim N$
(h) $\theta = 0.2, \gamma = 1.5$
$D(N)$
N
sims: n_1
theory: $n_1 \sim N^{0.5}$
sims: n_2
theory: $n_2 \sim N^{0}$